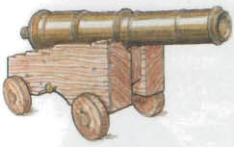


GUNS



FROM A BOOMING cannon to a pocket pistol, all firearms (guns) work on the same principle: a controlled explosion in one part of the gun propels a shell or bullet out of a tube or barrel. Firearms appeared in Europe in the early 14th century; although they were feeble at first, in time they changed warfare forever. Armour could not stop bullets, nor castle walls withstand cannon balls. Without guns, no nation could resist invasion for long; armed with them, European peoples colonized most of the world.



Howitzer in use

Artillery

A gun is described in terms of calibre, which is the width of the hole through its barrel, or the weight of the shell it fires. The shells of artillery (heavy firearms) are big enough to destroy buildings. Artillery includes mortars, with a fairly short range, or longer-barrelled, long-range guns.



Mountain howitzer

Hiram Maxim

American-born inventor Hiram Maxim (1840–1916) developed the first practical automatic machine gun in 1884. Maxim guns fired so quickly that by World War I (1914–1918) soldiers no longer fought on horseback, but tried to hide from the bullets in trenches.



Howitzers

A howitzer is a field gun, which may be towed or self-propelled to the battlefield. Its barrel and aim is midway between that of a mortar and a gun. Shells can be fired at a high angle, so that they fly above hills or other defences before reaching enemy targets.

Ammunition

The missile that a gun fires, the propellant (explosive charge), and the means of firing it, are known together as ammunition. The first guns fired round stones, propelling them from the barrel with loose gunpowder, lit with a hot wire. From the mid-19th century on, following the introduction of rifled gun barrels, shells were long and pointed in shape. Today, in all but the biggest guns, a metal cartridge holds the missile, propellant, and means of firing together.

Machine gun belt, World War I

250 rounds on the belt



Rifle cartridge



Used by 5.56-mm (.22-in) calibre gun

Machine gun bullet

Armour-piercing shell, fired by anti-tank gun.

Bullet, for .44 revolver

FIND OUT MORE

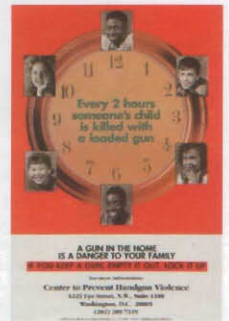


ARMS AND ARMOUR

Gun control

A few civilians need to own guns for security, target shooting, pest control, and other uses. However, guns are dangerous, and most countries control ownership. Many people believe these controls should be stricter to stop criminals getting hold of and using guns, and to reduce the number of shooting accidents.

Gun safety poster, USA



WEAPONS

WAREFARE

WORLD WAR I

15th-century illustration of a battlefield shows cannons in action.



Origins of guns

The first guns were cannons, known to have been in use before 1326, when drawings of them appeared in a book. At first smiths built the barrels from strips of iron. Safer, cast cannons came into use a century later, when bell-founders began to make them by filling a mould with liquid metal.

Small arms

Soldiers carry small arms for individual use. These compact, lightweight weapons may have a stock for bracing the gun against the shoulder, or they may be fired from the hand. Most fire bullets rapidly from preloaded magazines. Some are very powerful: a modern rifle can fire a bullet more than 1,800 m (2,000 yards).

.45-calibre bullet



Automatic Colt 1911 A1 pistol



Pistols

A pistol is easy to hide under clothing, and quick to fire. It has a short barrel and so cannot be aimed accurately at distant targets. It is most useful in close fighting or for self-defence.

Plastic hand guard

Submachine gun



Automatic weapons

Set to automatic, many small arms will continue to fire as long as the soldier holds back the trigger.

Sniper's rifle



Rifles

Rifling (spiral grooves) inside the barrel of a gun makes the bullet spin in flight, improving accuracy. Self-loading rifles use the energy from firing a bullet to expel the cartridge case, and load the next shot.

10-round magazine

Rifles are fired from the shoulder.